

Phase anomaly brings quantum implications

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Experimental confirmation that the Gouy phase can modify the photonic de Broglie wavelength opens up many exciting directions in metrology using quantum systems with higher-order Gaussian modes.

In 1905, to explain the photoelectric effect, Albert Einstein postulated that light sometimes behaves like individual particles rather than a wave. The discovery of these quanta of light – which became known as photons – earned him the Nobel Prize. Roughly two decades later, Louis de Broglie took the concept of wave–particle duality further and postulated that it is not only important for photons, but for all particles. For that, he introduced an effective wavelength – now called the de Broglie wavelength – that describes the quantum behaviour of general quantum systems. Commenting on de Broglie’s radical idea, Einstein remarked that “he has lifted a corner of the great veil” – and indeed this insight was critical for Erwin Schrödinger’s discovery of his famous wave equation for describing the wave function of quantum mechanical systems.

Fast forward 70 years, into the age of experimental quantum optics and entanglement research, scientists started to explore complicated multi-photon quantum states and their quantum interference patterns. It was subsequently discovered that the interference properties of a quantum state with N indistinguishable photons (each with a wavelength of λ_0) can be described in the same way as the interference of a single photon, but with an effective wavelength λ that is modified by dividing by N (ref. 1). This leads to the famous formula for the photonic de Broglie wavelength

$$\lambda = \frac{\lambda_0}{N}. \quad (1)$$

The photonic de Broglie wavelength has been observed in numerous experiments, including among others two celebrated works from 2004 using three indistinguishable photons² and a non-local four-photon state³. The scaling is not only interesting for fundamental reasons, but also has direct consequences for potential practical applications in quantum-enhanced measurement schemes. For example, imagine you want to measure a phase in an interferometer. If you are able to create a four-photon quantum system at a wavelength of $\lambda_0 = 800$ nm, your interference fringes will oscillate as a single photon with a wavelength of $\lambda = 200$ nm. The simplest way to understand this scaling is to think about each of the N indistinguishable photons accumulating the phase during propagation, leading to a phase of $e^{-iNkz} = e^{-iN2\pi z/\lambda_0}$. This effect and its derivatives are pivotal to the field of quantum metrology, and large efforts are under way to advance both the experimental and theoretical maturity, with the goal to develop applications for true practical situations⁴.

This scaling law, however, is not the final answer – as we learn from an article in *Nature Photonics* by Hiekkamäki, Barros, Ornigotti and Fickler from Tampere University, Finland⁵.

The Tampere team has shown that if we take into account the propagation of photons, there is another contribution that comes from the Gouy phase. The Gouy phase accumulates when a Gaussian beam propagates through a focus and was first observed by Louis Georges Gouy in 1890. It is somewhat of an unusual and unexpected phenomenon. Sometimes called the ‘phase anomaly’ it has intriguing interpretations such as a geometric⁶ or topological⁷ phase, and can be traced back to uncertainty relations in Gaussian beams⁸.

The authors have realized and experimentally confirmed that the Gouy phase modifies the celebrated photonic de Broglie wavelength in equation (1) to

$$\lambda = \frac{\lambda_0}{N(1+f(\lambda_0, \omega_0, z, m))} \quad (2)$$

where $f(\lambda_0, \omega_0, z, m)$ is a nonlinear function of the photon’s original wavelength λ_0 , the beam waist ω_0 , the distance from the focus z , and the transverse spatial mode order m , and contains the accumulated Gouy phase Φ_G . This modification can be seen from the additional phase contribution from the Gouy phase. The total phase due to the propagation then becomes $e^{-iNkz - iN\Phi_G(\lambda_0, \omega_0, z, m)}$, where $\Phi_G(\lambda_0, \omega_0, z, m) = -m \times \arctan\left(\frac{z\lambda_0}{\pi\omega_0^2}\right)$ is a nonlinear function in λ_0 , ω_0 and z (Fig. 1a).

The Gouy phase itself can be derived from pure classical wave mechanics. However, its highly non-trivial effect on modifying the photonic de Broglie wavelength is a clear vindication of its ‘quantumness’, an insight that took a surprising 27 years to unfold (since ref. 1).

Now let us dive deeper into how Hiekkamäki and colleagues were able to observe this phenomenon. To observe this effect, the authors create so-called NOON states, where you have N photons in mode a and zero in mode b , or vice versa, that is $|N\rangle_a |0\rangle_b - |0\rangle_a |N\rangle_b$. In their experiment, modes a and b are the orthogonal transverse spatial modes in Laguerre Gauss modes. Specifically, the authors chose a to be the fundamental Gaussian mode and b as the radial mode number p (which takes values from 1 to 4 in different experiments). The Gouy phase depends linearly on the radial modes and therefore modifies the de Broglie wavelength. The experiment is performed with two-photon states, that is $N = 2$, such that the quantum state used in their experiment can be read as $|\phi\rangle = |2\rangle_0 |0\rangle_p - |0\rangle_0 |2\rangle_p$. In the experiment, they use a 405 nm continuous-wave laser to pump a nonlinear crystal such that photon pairs at 810 nm are produced via spontaneous parametric down-conversion – a popular approach for generating entangled photon pairs. The photon’s transverse spatial wavefront is modulated independently at two regions of one spatial light modulator. Each photon is transformed into a cleverly constructed superposition of a fundamental Gaussian mode and a higher-order radial mode. The two photons then propagate to a beam splitter and undergo Hong–Ou–Mandel interference. In that way, the two photons can bunch into the same beam path such that a radial mode NOON state appears (Fig. 1b).

The idea to observe the quantum effect of the Gouy phase is to measure the probability of two-photon states along the beam axis. This

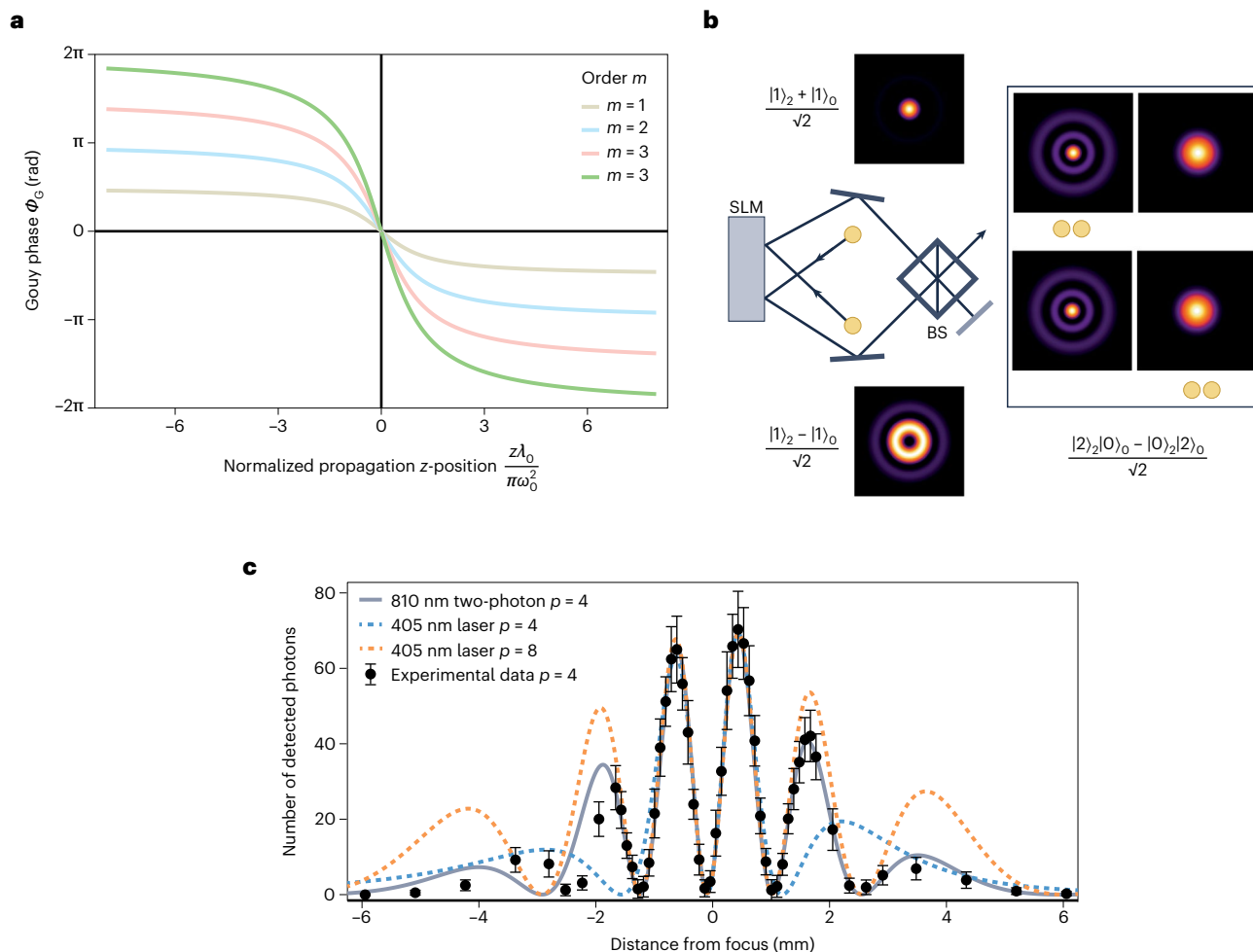


Fig. 1 | The classical and quantum Gouy phase. **a**, The Gouy phase shift $\Phi_G(\lambda_0, \omega_0, z, m)$, which is well-known in classical optics for Gaussian beams, along the propagation direction for a beam with wavelength λ_0 , beam waist ω_0 , distance from focus z and transverse spatial mode order m . **b, c**, Two individual photons are encoded into spatial superpositions (the ket stands for photon numbers, the subscript for the radial mode number) (**b**). When they bunch at a

beam splitter (BS), they produce a radial NOON state, which is used to observe the genuine quantum effect of the Gouy phase – namely a modification of the photonic de Broglie wavelength, which results in the quantum interference fringes in **c**. SLM, spatial light modulator. Panel **c** reproduced from ref. ⁵, Springer Nature Ltd.

is possible as the Gouy phase depends on the distance from the focus and the size of the focus. To observe its effect in quantum interference, the authors focus the NOON state to a 25-micrometre-sized Gaussian waist, and use a translation stage with a single-mode fibre to collect photons. For each position, they collect the number of two-photon events. The observed two-photon count rate is well described by their numerical computation, and cannot be described by a laser with half of the wavelength (even if the mode number is adapted too) (Fig. 1c). This result clearly establishes a new quantum interference behaviour that plays a crucial role in quantum entangled states consisting of indistinguishable multiphoton states with complex spatial structures.

While these results are of course exciting for the fundamental reason that they transmute the decade-old photonic de Broglie wavelength, the result also has clear potential for applications. The quantum Gouy phase evolves faster for a larger number of photons

and a larger mode number. Therefore, one could use this effect as a quantum-enhanced technology for sensitive measurements of positions in longitudinal directions. One interesting advantage of using radial NOON states (rather than the more common NOON states where photons can be in two different paths) is that the photons co-propagate, thus are quasi intrinsically phase stable and robust against fluctuations, which is crucial for practical applications.

Looking into the future, it is likely that these new features that emerge from the quantum effects of the Gouy phase will become a standard component of the toolbox of quantum metrology researchers. The complex and nonlinear dependence of the de Broglie wavelength in equation (2) on the photon wavelength, the mode order, distance from the focus, and beam waist leaves a lot of room for exploring new consequences theoretically and experimentally. Further opportunities may also arise from the use of more complex photon-number

entangled quantum states, such as platonic solid states, which are advantageous for certain quantum metrology tasks. Conceptually, it would also be interesting how this phenomenon translates to the non-paraxial regime of strongly focused beams, which might be relevant for quantum-enhanced microscopes.

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Competing interests

The authors declare no competing interests.